DeepSeek Harness: Everything is a plugin

Master view E0

Every part is a plugin, including the agent loop, the model adapter and the tool registry, so each can be replaced from configuration.
The web server binds to your own machine, and every request speaks for one operator.
The project calls itself a developer preview that has not had a security audit.

Composed at launch

W8 W9

Later layers win, row by row

- **Bundle patches** — In the profile bundle order
- **Profile patch** — Your layer for one profile
- **Home patch** — Applies to every profile
- **Command line overlays** — Patch files in argument order
- **Telemetry switch** — Applied last

Plugin tree — Profile bundles plus patch layers

Plugin lifecycle

W8

Install to unload, the tree changes while it runs

- **Install a bundle** — pnpm add in the profile folder — pnpm 11.7.0
- **Peer check** — Rejects a plugin the runtime cannot satisfy
- **Load into the tree** — Loader imports and mounts it — Node 22.19 or 24
- **Hot reload** — Config changes apply without restart
- **Unload undoes effects** — Every registration is reversible

Machine boundary

Callers

W6 W10

Each launches a named profile

- Web app → Web profile
- Desktop app → Desktop profile
- One shot command line → Headless profile
- SDK programs — TypeScript and Python over stdio → SDK profile / SDK minimal profile
- ACP clients — Editors and automation → ACP profile

From the page to the host

- **Browser page** — React single page app, built by Vite
- **Remote calls** — HTTP POST per namespace and method
- **Stream socket** — One WebSocket carries many streams
- **Web server** — HTTP and upgrade routes
- **Client connection** — Token, cookie and host checks
- **On your own machine** — Loopback web server, one operator
- **Desktop host** — Electron in Node mode, port 19387

The harness

Stated design rules: Everything is a plugin · Model visible means logged · Plugins, not loop changes · Misconfiguration fails loud · Explicit over implicit

W0 W1

One Node host process on your machine, built from plugins on Cordis 4.0.4

Presets — W7

- Standard mode
- PTC mode
- Minimal mode
- Creator mode
- Agent Teams

Compaction — W2

When the context fills up

- **Pressure trigger** — At 80 percent of the window
- **Recent tail kept** — Last 16 percent kept verbatim
- **Summary logged** — Replaces the compacted range

Should compact? · Summary logged

Model request path — W11

One send, from build to log

- Build request
- Resolve route
- Send the request
- Stream the reply
- Retry on failure
- Meter tokens
- Append to session log
- Credentials lookup

Model calls over HTTPS

Delegation — W7

Child agents, up to 8 at once, one level deep by default

- Spawn
- Fork
- Workflow engine
- Codex — Off
- Claude Code — Off
- ACP subagent — Off
- Active children
- Delegation depth
- Messages to a child

Delegates

1 Inbox — W1

- **Follow up message** — Queues the next turn prompt
- **Steering input** — Joins the next step of the turn
- **Injected context** — Waits for the next admitted step
- **Webhook delivery** — Starts a new session per delivery — Off

Input

2 Model context — W2

- System prompt node
- Runtime context
- Skills and instructions
- File references
- Session references
- Model history

Prompt

3 Agent loop — W1 W11

One model call and its tools per step

Model service — Provider neutral adapter seam

Tool calls → **Tools** — Shell, files, search, web, skills

Results →

- **System prompt** — Prompt sections and tool schemas
- **Model selection** — Default model, chosen per session
- **Stream the reply** — Shown live, logged when complete
- **Parallel tool calls** — Up to 10 at once in a step

Each call

4 Tool gate — W1 W4

Allowed?

- Pre execute
- Approval
- Guards

Allowed

5 Execution world — W3 W6

Tools run here, confined by the sandbox

- Shell command
- Terminal tools — Off
- Web search
- Read, write and edit
- Plan, goals and todos
- Web fetch
- Glob and grep
- Ask the user
- MCP client — Off
- Run code
- Load a skill
- Browser and computer use — Off
- Background jobs
- Office documents — Off
- Remote over SSH — Off

Result logged, back to the model

6

Session log

Append only, one row per event, the source of model history

W2

- **Turns and steps** — Open and close markers, with a reason
- **Request headers and context**
- **Messages** — System, developer, user and assistant
- **Failed attempts** — Kept in the log, never model history
- **Plugin events** — Approvals, compaction, subagent events
- **Tool calls and results** — Each call paired with its outcome
- **Session search** — Full text, past sessions — Off
- **Versioned JSONL file** — One row per event, version in the name
- **Compressed frames** — Zstandard by default, or plain lines
- **Write lease** — One writer, flock or a named semaphore
- **Atomic publish** — Temp write, fsync, then publish
- **Session projections** — Typed state folded from the log
- **Key value storage** — JSON files for settings and state
- **Attachments** — Content addressed files

Model providers

W11

Reached over HTTPS

- **DeepSeek API** — API key route, the default — deepseek flash
- **DeepSeek account** — Browser sign in route
- **Compatible endpoints** — OpenAI or Anthropic style APIs
- **Your own endpoint** — Custom base URL and key
- **Anthropic messages** — Anthropic Messages route
- **OpenAI completions** — Chat completions route

Other external services

W3

Reached by tools and installs

- **Web search providers** — DeepSeek by default, Exa, Perplexity
- **Exa search** — Opt in provider, api.exa.ai — Off
- **MCP servers** — stdio or Streamable HTTP — Off
- **Perplexity search** — Opt in provider, api.perplexity.ai — Off
- **GitHub** — Webhook deliveries — Off
- **npm registry** — Plugin and bundle installs
- **Codex and Claude Code** — External agents as subagents — Off
- **Reference memory** — Knowledge graph, JSONL file — Off

Evaluated and shipped

W5 W8

Proved before release, watched while it runs

After feedback

Before it merges

- Unit tests
- Coverage gate
- Snapshot replay
- Pull request checks
- Sandbox proofs

Releases

- npm packages
- Python wheels
- Desktop installers
- Update feed
- Signed builds

While it runs

- Session telemetry
- Redaction step
- Token meter
- Runtime invariants — Off

Security and trust

W4

What keeps the loop in check

Enforce · Limit

Access and secrets

- Loopback bind
- Launch token
- Signed cookie
- Credential references
- Account sign in

Guards and records

- Write fence
- Repeat call reminder
- Tool timeout
- Environment scrub
- Every step in the session log

Approval

- Web approval card
- Four outcomes
- Workspace preset
- Full access preset
- Auto review

Sandbox

- Workspace write — bubblewrap
- Read only — Landlock
- Full access — Seatbelt
- No usable backend — Restricted token

Capability seams

S1 S1B

33 seams, each one swappable

- Model service
- Session persistence
- Sandbox
- Approval
- Shell
- File system
- Subagents
- Web search and fetch

Page guide

Why each page exists

- W0 **Reference architecture:** every part of the web profile on one sheet
- W1 **Agent harness internals:** one turn, the tool pipeline, the core services
- W2 **Session log, context and memory:** what is logged, stored and sent to the model
- W3 **Tools and integrations:** what the model can call, and which systems it reaches
- W4 **Security, trust and governance:** sandbox, approval, credentials, web access
- W5 **Observability and evaluation:** runtime signals, tests, benchmarks and CI
- W6 **Runtime topology and decisions:** which profile, which process, which sandbox
- W7 **Agent catalog and worked flow:** presets, delegation, one approval traced
- W8 **Lifecycle and operating model:** composition order, plugin lifecycle and releases
- W9 **Profile and bundle composition:** what each profile and agent preset mounts
- W10 **Client and host topology:** browser page, host process, desktop shell, SDKs
- W11 **Model layer:** how a request reaches DeepSeek, and where the model is chosen
- S1 **Capability seam map:** each swappable seam, its providers and its consumers
- S1B **Seam map, continued:** the remaining seams
- S2 **Legend and citation register:** how to read every mark on these pages

DeepSeek Harness reference architecture

The web profile as built: one host process on your machine, composed from plugins

W0

Machine boundary

JSON RPC

Callers

People and programs

- C001 **Web app** — Browser UI on port 3080
- C002 **Desktop app** — Electron 44
- C003 **One shot command line**
- C004 **SDK programs** — TypeScript and Python over stdio
- C017 **ACP clients** — Editors and automation

DeepSeek Harness host

Cordis 4.0.4

One Node process on your machine, built from plugins

- C018 **Plugin tree**
- Remote — C019 **Web server and API gateway**

Agent harness

Cordis services

Detail on w1

- C005 **Agent loop** — One model call and its tools per step
- C020 **Agent registry** — Live agents and their events
- C022 **Model selection** — Default model, chosen per session
- C021 **System prompt** — Prompt sections and tool schemas
- C023 **Context compaction** — Summarizes when context fills up
- C006 **Tool registry** — Guarded pipeline for every call
- C007 **Subagents** — Delegates work to child agents
- C024 **Workflow engine** — Scripts that run many subagents
- C025 **Plan, goals and todos** — Plan mode, session goals, todo list

Session and context

- C008 **Session log**
- Off C026 **Session search**
- C027 **Skills and instructions**

Execution world

Cordis services

Detail on w6

- C028 **Shell**
- C029 **Subprocesses**
- C030 **File system**
- C031 **Code runtime**
- C032 **Background jobs**
- C033 **Web search and fetch**
- C034 **Terminal tabs**
- Off C035 **Language servers**
- Off C009 **Remote over SSH**
- C036 **Browser and computer use**

Agent presets

YAML patches

Detail on w7

- C037 **Standard mode** — Default, the full tool set
- C038 **PTC mode** — Tools called in batches from code
- C039 **Minimal mode** — Only a terminal tool
- C040 **Creator mode** — Agent writes plugins for DSH
- C041 **Agent Teams** — Lead, teammates, task board

Protocols and storage

How the host connects

Caller protocols

- C042 **Remote API**
- Off C043 **SDK server**
- Off C044 **ACP server**

Model adapters

- C045 **DeepSeek adapter**
- C046 **pi ai adapter**
- C047 **Retry policy**

Tool protocols

- Off C010 **MCP client**
- Off C048 **Hook bridges**
- Off C049 **GitHub webhook**

Local storage

- C050 **Session files**
- C051 **Key value storage**
- C052 **Attachments**

MCP

Model calls

MCP

Model providers

Reached over HTTPS, detail on w11

- C011 **DeepSeek API** — API key route, the default — deepseek flash
- C053 **DeepSeek account** — Browser sign in route
- C012 **Compatible endpoints** — OpenAI or Anthropic style APIs
- C054 **Your own endpoint** — Custom base URL and key

Other external services

Used by tools and installs, detail on w3

- Off C055 **MCP servers** — stdio or Streamable HTTP
- C013 **Web search providers** — DeepSeek by default, Exa, Perplexity
- Off C056 **GitHub** — Webhook deliveries
- C057 **npm registry** — Plugin and bundle installs
- Off C058 **Codex and Claude Code** — External agents as subagents

Trust

Security, observability and governance

- C014 **Sandbox**
- C059 **Write fence**
- C015 **Approval**
- C060 **Permission presets**
- C061 **Loop guards**
- C062 **Credential store**
- C063 **Launch token and cookie**
- C064 **Session telemetry**
- Off C065 **Runtime invariants**

Stated design rules

- Everything is a plugin
- Model visible means logged
- Plugins, not loop changes
- Misconfiguration fails loud
- Explicit over implicit

Shipped on by default

Off **Shipped** off by default or opt in

Experimental experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP protocols and trust links

As built, DeepSeek Harness 0.1.7 rc.2, source snapshot 25 September 2026

Agent harness internals

One turn from input to logged result, and the services every agent reuses

Back to master | W1

Where a turn starts

One inbox feeds the loop

- C067 **Follow up message** — Queues the next turn prompt
- C068 **Steering input** — Joins the next step of the turn
- C069 **Injected context** — Waits for the next admitted step
- Off C070 **Webhook delivery** — Starts a new session per delivery

One turn, step by step

A step is one model call plus its tools

Agent loop

1. C071 **Turn start** — Logged before input is claimed
2. C072 **Claim input** — Next step input plus one queued message
3. C073 **Assemble prompt** — Sections, tool schemas, runtime context
4. C074 **Pre step decision** — Listeners enter, rewrite or reject
5. C075 **Prepare the call** — Route resolved before anything commits
6. C076 **Freeze history** — Model history derived from the log
7. C077 **Stream the reply** — Shown live, logged when complete
8. C078 **Run tool calls** — Each call through the tool pipeline
9. C079 **Step end** — Next step when tools owe a request
10. C080 **Turn stopping** — Listeners may still stop the turn

Model calls

Tool pipeline

Every call, in this order

Tool registry

1. C081 **Call logged** — Recorded before it runs
2. C082 **Pre execute** — Hooks may allow, deny, cancel or ask
3. C083 **Approval** — Only allowed once goes ahead
4. C084 **Guards** — Can only deny, never allow
5. C085 **Execute wrappers** — Timeouts and checkpoints around the body
6. C086 **Post execute** — Accept, block or replace the result
7. C087 **Result logged** — One outcome the model sees

Model call

Detail on w11

- C091 **Request series** — Fresh header when the prompt changes
- C047 **Retry policy** — Empty response, rate limit, server, timeout, transport
- C092 **Cancellation** — Live, commits nothing mid call
- C093 **Token meter** — Context pressure from logged usage

Usage read

Core services

Built once, reused by every agent

Cordis plugins

- C005 **Agent loop** — One model call and its tools per step
- C020 **Agent registry** — Live agents and their events
- C021 **System prompt** — Prompt sections and tool schemas
- C006 **Tool registry** — Guarded pipeline for every call
- C008 **Session log** — Append only record of every step
- C088 **Model service** — Provider neutral adapter seam
- C089 **Scoped registration** — Scope tags, scope filtered events
- C090 **Session projections** — Typed state folded from the log

Extension points

Where new behavior attaches

Cordis 4.0.4

- C094 **Session events** — Durable facts, survive a reload
- C095 **Agent events** — Observe or intercept work in flight
- C096 **Capability events** — Policy on files, tools, telemetry
- C097 **Waterfalls call next** — Returning early stops the chain — Cordis

Limits and guards

Keeps a runaway loop in check

- C098 **Repeat call reminder** — After 3, 5 and 8 identical calls
- C099 **Tool timeout** — Per tool deadline, returns a timeout
- C100 **Parallel tool calls** — Up to 10 at once in a step
- C101 **Durability checkpoints** — Log flushed before calls and side effects

A step is one model request plus the tools it calls. A turn is zero or more steps.

Anything that reaches a model request must be rebuildable from the session log.

New behavior goes on documented extension points. Changing the loop means updating the architecture doc.

A failed model call is retried inside the same open turn, over the same logged history.

Blocks follow one agent turn in order. Event names for each step are listed in the citation register.

Shipped — on by default

Off Shipped — off by default or opt in

Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

As built, DeepSeek Harness 0.1.7 rc.2, source snapshot 25 September 2026

Session log, context and memory

The append only log, how it is stored and read, and what reaches the model

What the log records

Append only, one row per event

- C102 **Turns and steps** — Open and close markers, with a reason
- C103 **Messages** — System, developer, user and assistant
- C104 **Tool calls and results** — Each call paired with its outcome
- C105 **Request headers and context** — Route and envelope per request series
- C106 **Failed attempts** — Kept in the log, never model history
- C107 **Plugin events** — Approvals, compaction, subagent events

On disk

The only first party backend

- C108 **Session folder** — Under DSH_HOME, grouped by project
- C109 **Versioned JSONL file** — One row per event, version in the name — JSONL, Version 4
- C110 **Compressed frames** — Zstandard by default, or plain lines — Zstandard, node:zlib
- C111 **Write lease** — One writer, flock or a named semaphore — flock
- C112 **Atomic publish** — Temp write, fsync, then publish

Format versions

Adjacent migrations, one package each

1. C113 **Version 0 to 1** — Identity migration
2. C114 **Version 1 to 2** — Assistant stream migration
3. C115 **Version 2 to 3** — System prompt and envelope
4. C116 **Version 3 to 4** — Tool role migration, current writer

Reading the log

Derived views, never a second store

- C117 **Model history** — Derived from the log each step
- C090 **Session projections** — Typed state folded from the log
- C118 **Projection cache** — Checkpoint every 200 events or 5 s
- C119 **Session stats** — Counts and wall times
- C120 **Turn outline** — Whole log outline per turn

Context the model sees

Every input comes from the log

- C121 **System prompt node** — Logged as a system message
- C122 **Runtime context** — Per step snapshot beside the prompt
- C027 **Skills and instructions** — Skill folders and AGENTS.md files
- C123 **Session references** — Snapshots of other sessions
- C124 **File references** — At mentions of workspace files
- Off C125 **Time context** — Current and elapsed time

Compaction

When the context fills up

- C126 **Pressure trigger** — At 80 percent of the window
- C127 **Overflow recovery** — Compacts after a context overflow
- C128 **Compact command** — On demand, from a slash command
- C129 **Recent tail kept** — Last 16 percent kept verbatim
- C130 **Tool result pruning** — Model free head and tail trim
- C131 **Image offload** — Placeholders for over budget images
- C132 **Summary logged** — Replaces the compacted range

Large tool output

- C133 **Spill policy** — Inline up to 12500 tokens
- C134 **Spill store** — Private files in the OS temp folder

Other stores

Beside the session log

- C051 **Key value storage** — JSON files for settings and state
- C052 **Attachments** — Content addressed files
- Off C026 **Session search** — Full text, past sessions — SQLite FTS5
- Off C135 **SQLite storage** — Other key value backend — SQLite, node:sqlite
- C404 **Content hash names** — SHA 256 hex digest as the filename — SHA 256
- C136 **Session export** — Export command and download

Memory through MCP

MCP servers

Reference configs, off by default

- Off C137 **Memorix** — Reference config, stdio
- Off C138 **Reference memory** — Knowledge graph, JSONL file
- Off C139 **Engram** — Reference config, stdio

The session log is the source of the context the model sees. Model history is derived from it.

Committed files are never renamed, replaced or deleted. A write publishes the next version beside them.

Compaction is logged like everything else, as a summary that replaces the compacted range.

Large tool output is saved to private files and the model gets a way to read it back.

Session search indexes past sessions with SQLite FTS5 virtual tables, off by default.

Fourteen core event types, plus plugin events such as approvals and compaction. Names are in the register.

Shipped — on by default

Off Shipped — off by default or opt in

Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

Tools and integrations

What the model can call, and how the harness reaches other systems

Work in the workspace

Files, commands and code

- C140 **Shell command** — A process per call — bash, PowerShell
- C141 **Read, write and edit** — Also reads images
- C142 **Glob and grep** — Packaged ripgrep — ripgrep 1.18.0
- C143 **Run code** — PTC mode, programs call the tools
- C144 **Persistent shell** — Minimal mode, one PTY terminal — bash, PowerShell
- Off C145 **Terminal tools** — Open, send, read and close PTYs — node pty, xterm headless
- C405 **Background jobs** — List, read output, or kill a job
- Off C406 **Office documents** — SDK app only — LibreOffice Kit 0.1.1

Plan and coordinate

Keep the user in the loop

- C146 **Exit plan mode** — The plan goes to the user for review
- C147 **Goals** — Create, read and update a goal
- C148 **Todo list** — Written to the session log
- C149 **Ask the user** — Pauses until the user answers
- C150 **Present files** — Declares workspace deliverables
- C151 **Load a skill** — Reads a skill on demand
- Off C407 **Schedule a message** — After a delay, at a time, or cron
- C408 **Inspect the plugin tree** — Creator mode, read only

Delegate

Detail on w7

- C152 **Subagent** — A continuable child agent
- C153 **Fork** — A child seeded with this history
- C154 **Message and interrupt** — Steer, stop or list children
- C155 **Workflow** — A script that runs many subagents
- Off C156 **Ralph loop** — Fresh agents in a loop
- Off C409 **Codex subagent** — One shot, the app server protocol
- Off C410 **Claude Code subagent** — One shot, the Agent SDK

Look things up

Web, resources, history, code

- C157 **Web search** — DeepSeek search by default
- Off C411 **Exa search** — Opt in provider, api.exa.ai
- Off C412 **Perplexity search** — Opt in provider, api.perplexity.ai
- C158 **Web fetch** — Public pages, same origin redirects
- C159 **MCP resources** — List and read server resources
- Off C160 **Session history** — Search and trace past sessions
- Off C161 **Code navigation** — Definitions, references and hover

MCP client

One row per server, none by default

- Off C010 **MCP client** — Connects configured MCP servers — MCP SDK 2.0.0
- Off C162 **Two transports** — stdio or Streamable HTTP — MCP SDK 2.0.0
- Off C163 **Server qualified names** — Server name plus the tool name
- Off C164 **Server instructions** — Join the logged system prompt

Other protocols

Opt in, each its own package

- Off C044 **ACP server** — Separate acp profile, stdio
- Off C165 **ACP subagent** — Drives an ACP agent as a child — ACP SDK 1.4.0
- Off C043 **SDK server** — Separate sdk profile, stdio
- Off C048 **Hook bridges** — Claude Code and Codex hook files
- Off C049 **GitHub webhook** — Starts a session per delivery

When a tool asks first

Detail on w4

- C174 **Wider sandbox request** — A retry with a justification asks
- Off C175 **Plugin manager tool** — Full access or approval per call
- Off C176 **Hook decisions** — Pre tool hooks may block or ask

Browser and computer use

Experimental providers

- C166 **Playwright MCP** — Chromium through Playwright — Playwright MCP 0.0.80
- C167 **Chrome DevTools MCP** — Chromium through DevTools — Chrome DevTools MCP 1.9.0
- C168 **Stagehand** — Browser tools with own models — Stagehand 4.1.0
- C169 **Cua driver** — Desktop computer use — cua driver 0.28.0

Remote over SSH

Same tools, another host

- Off C170 **SSH connection** — OpenSSH plus a remote helper
- Off C171 **Remote files** — File system on the remote host
- Off C172 **Remote processes** — Subprocesses and terminals
- Off C173 **Remote sandbox** — Confines commands remotely

In the web profile, tools come from the chosen agent preset. The host rows for them are switched off.

MCP tools are named after their server, so session history and permission rules survive restarts.

Web search defaults to the DeepSeek provider, reached through the DeepSeek API.

Over SSH only files, processes and the sandbox move. The harness, model calls and sessions stay local.

Tool names for every block are listed in the citation register. Seventy tool entries ship in the repo.

Shipped — on by default · Off Shipped — off by default or opt in · Experimental — experimental package, opt in · MCP · ACP · JSON RPC · Remote · Trust · OTLP — protocols and trust links

Security, trust and governance

Sandbox, approval, credentials, web access and what leaves the machine

Sandbox modes

One mode per session

- C177 **Read only** — Commands may not write files
- C178 **Workspace write** — Default: workspace only
- C179 **Full access** — Skips the sandbox entirely
- C180 **No usable backend** — Fails closed with an error

Sandbox backends

The code really splits by platform

Linux — *Linux kernel*

- C181 **bubblewrap** — Linux, the first choice — `bubblewrap`
- C182 **Landlock** — Linux fallback, native launcher — `Landlock`

macOS — *Seatbelt*

- C183 **Seatbelt** — macOS sandbox exec profile — `Seatbelt`

Windows — *Restricted token*

- C184 **Restricted token** — Windows, always partial — `ACL`

SSH remote — *SSH*

- Off C173 **Remote sandbox** — Confines commands remotely

The sandbox confines what commands may write. Each call reports whether enforcement was full or partial.

Full access skips the sandbox, and the approval policy then becomes never.

An unanswered approval defaults to unavailable, which denies the call.

Every approval is written to the session log as an asked and decided pair.

Official DeepSeek requests carry a session log suffix by default, and telemetry is sent only after feedback.

Approval

One request() call, in this order — *User approval*

1. C413 **Ask logged** — Before any channel answers
2. C414 **Never policy** — Rejects without asking anyone
3. C415 **Routed to a channel** — Web card or ACP answers
4. C186 **Four outcomes**
5. C416 **Decision logged** — Paired with the ask by its id

Approval channels

Whichever channel the route picks

- C189 **Web approval card** — Allow once or reject
- Off C190 **ACP client answers** — Through its permission request

Permission presets

Sandbox mode plus approval policy

- C192 **Read only preset** — Read only sandbox, asks first
- C193 **Workspace preset** — The default, asks first
- C194 **Full access preset** — No sandbox, never asks
- C195 **Auto review** — An LLM reviews each tool call

Credentials and identity

Secrets stay out of settings

- C196 **Credential references** — Settings name a secret, never hold it
- C197 **Lookup order** — Environment, credentials file, then .env
- C198 **Owner only file** — Mode 600 in a 700 folder
- C199 **Account sign in** — Browser PKCE with S256
- C200 **Anonymous id** — Random id for telemetry and feedback

Web access control

Who may call the host

- C201 **Loopback bind** — All interfaces refused by the CLI
- C202 **Launch token** — One per process, in the first URL
- C203 **Signed cookie** — HttpOnly, SameSite Strict, 30 days
- C204 **Host and origin checks** — Refuses rebinding and cross site calls
- C205 **Body size cap** — 300 MiB per API request

Sent as a header

Process and file hygiene

- C206 **Environment scrub** — Secret named variables kept from children
- C059 **Write fence** — File tools stay in the workspace
- C185 **Read before edit** — Edits need a fresh read
- C061 **Loop guards** — Repeat reminders and timeouts

What leaves the machine

Outbound data, mostly on by default

- C207 **Model requests** — To the configured provider
- C208 **Session log in requests** — Up to 8 MiB added to DeepSeek calls
- C209 **Telemetry after feedback** — OTLP logs, opt out by environment
- Off C210 **Product telemetry** — Opt in plugin, explicit events

Stated safety posture

In the project's own words

- C211 **Developer preview** — Breaking changes expected
- C212 **Not security audited** — Stated in the safety notice
- C213 **Run it contained** — A disposable VM or container

Safety posture comes from the project README and SAFETY.md. Every other block cites code or config.

Shipped on by default · Off **Shipped** off by default or opt in · **Experimental** experimental package, opt in · MCP · ACP · JSON RPC · Remote · Trust · OTLP protocols and trust links

Observability and evaluation

What the harness reports while it runs, and how quality is proved before release

Session telemetry

Detail on w4, what leaves the machine

- C064 **Session telemetry** — Sent only after user feedback — OpenTelemetry JS 0.220
- C214 **OTLP log export** — Gzip logs to DeepSeek telemetry — OpenTelemetry JS 0.220
- C215 **Two modes** — Feedback only, or disabled
- C216 **Redaction step** — Rules can rewrite a record before export
- Off C210 **Product telemetry** — Opt in plugin, explicit events

Feedback and export

Human signals, in the log

- C217 **Feedback command** — Session feedback, logged only
- C218 **Message ratings** — Ratings and notes on replies
- C136 **Session export** — Export command and download
- C208 **Session log in requests** — Up to 8 MiB added to DeepSeek calls
- C209 **Telemetry after feedback** — OTLP logs, opt out by environment

Runtime checks

While it runs

- Off C065 **Runtime invariants** — Checks the log rebuilds requests
- Off C219 **Request rebuilt from log** — Invariant checks each model call
- C093 **Token meter** — Context pressure from logged usage
- C220 **Console logger** — Vendored Cordis logger
- C221 **Inspector** — DevTools hub for host debugging

Test tiers

Before anything merges

- C222 **Unit tests** — Vitest across every package — Vitest 4.1.8
- C223 **Coverage gate** — 100 percent per file in package sources
- C224 **Real API tests** — Skipped without an API key
- C225 **Snapshot replay** — Recorded sessions, no key needed
- C226 **Lint, types, duplication** — Static checks before any test

Benchmark gates

Six performance gates

- C227 **Stream reconnect** — Active assistant reconnect
- C228 **Agent continuation** — Backend continuation
- C229 **Conversation fold** — Long chat folding in the client
- C230 **Long session browser** — Built client in a real browser
- C231 **Session open** — Opening stored sessions
- C232 **Terminal IO** — Persistent terminal throughput

CI on GitHub

Workflows in the repo

- C233 **Pull request checks** — Static, coverage, benchmarks, snapshots
- C234 **Node version matrix** — Node 22.19 and 24.9 and newer
- C235 **Live API run** — On pull requests and nightly
- C236 **Sandbox proofs** — Real kernel runs of each backend
- C237 **macOS and Windows lanes** — Serial runs after merge
- C238 **Weighted review** — Approval score before merge

Repository gates

Scripted checks

- C239 **Runtime closure** — Shipped packages resolve cleanly
- C240 **Cordis config check** — Bundle rows name real plugins
- C241 **Module graph** — Package layering checked
- C242 **Docs sync** — Catalogs and links match the code

Postmortems

Written up in the docs

- C243 **ACP default export** — Lost the inject method
- C244 **Expression disabled file tools** — A config literal switched them off
- C245 **Web agent feedback loop** — The agent validated the wrong server
- C246 **Landlock partial notice** — Child failures misclassified

Telemetry releases the session log only after explicit user feedback, and an environment variable turns it off.

Coverage is gated at 100 percent per file for package sources.

Every non trivial user visible change updates a recorded session snapshot.

Benchmark timing runs in workers under plain Node, against built packages.

Runtime signals come from shipped rows. Test, benchmark and CI blocks cite the workflow and script files.

Shipped — on by default

Off Shipped — off by default or opt in

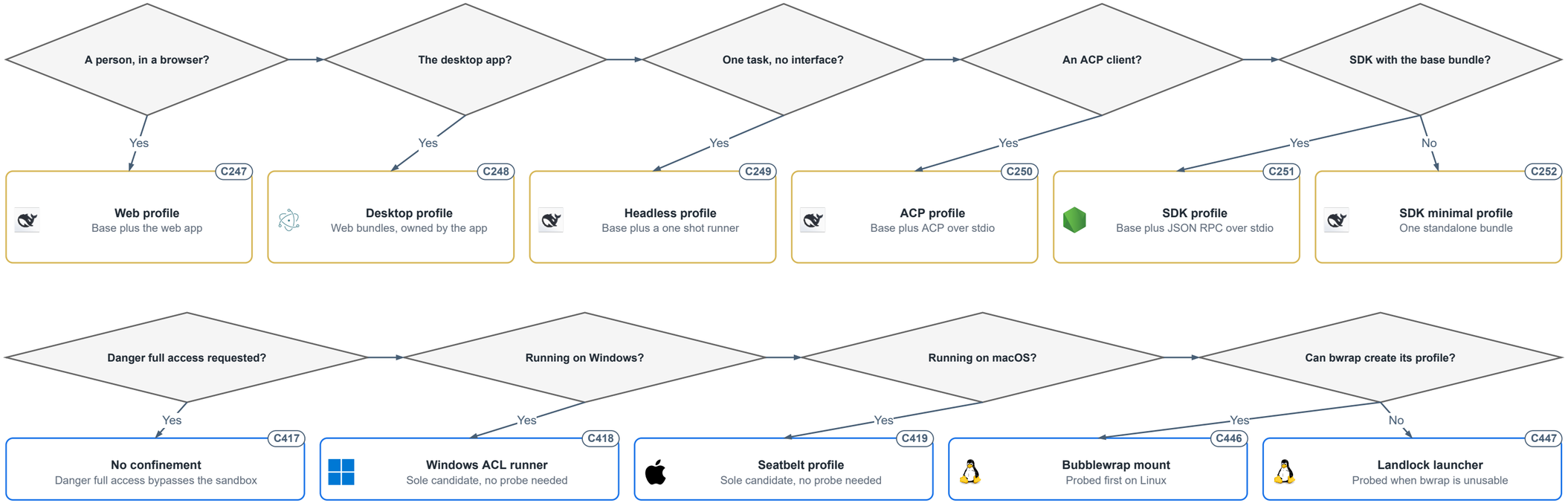
Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

Runtime topology and decisions

Which profile to launch, what runs in which process, and where data lives

Back to master | W6



Every supported Node application launches through a named dsh profile.

Desktop runs the same web stack in its own host process, with its own reserved profile.

File and subprocess providers share one execution world, so a remote provider moves shells, terminals and language servers together.

The SDK and ACP profiles run the same dsh program and talk over stdio.

A platform chain is probed in preference order only when it has more than one candidate. Linux tries bubblewrap before the Landlock launcher, macOS and Windows have a sole candidate each.

When no backend on the host is usable, a confined mode fails closed instead of running the command unconfined.

Processes

What runs where

- C253 **dsh launcher** — Node 22.19, or 24 and later
- C254 **Host process** — Runs the composed plugin tree
- C255 **Browser page** — React single page app, built by Vite
- C256 **Electron main** — Windows, app scheme, updates, tray
- C257 **Desktop host** — Electron in Node mode, port 19387
- C258 **Child processes** — Shells and code runtime

Data home

Shared by the CLI and Desktop

- C259 **Data root** — DSH_HOME, else .dsh in the home folder
- C260 **Profiles folder** — Bundles, patch file, installed plugins
- C050 **Session files** — One active file, prior versions kept
- C051 **Key value storage** — JSON files for settings and state
- C261 **Credentials file** — Owner only, beside the profiles
- C052 **Attachments** — Content addressed files

One execution world

Swap the providers and everything moves

- C030 **File system** — Read, write, edit, search
- C029 **Subprocesses** — Process groups, bounded output
- C028 **Shell** — Bash or PowerShell, confined
- C031 **Code runtime** — Model written programs in Node
- Off C035 **Language servers** — Definitions and references
- Off C170 **SSH connection** — OpenSSH plus a remote helper

Platforms and packaging

Detail on w8

- C262 **Desktop installers** — macOS DMG and ZIP, Windows NSIS
- C263 **Python runtime wheels** — Linux, macOS 14 and Windows x64
- C264 **Native addon** — Landlock launcher and file locks

Profiles and bundle lists come from the profile templates in app boot. Detail on w9.

Shipped on by default | Off **Shipped** off by default or opt in | **Experimental** experimental package, opt in | MCP ACP JSON RPC Remote Trust OTLP protocols and trust links

Agent catalog and worked flow

Four agent presets, how work is delegated, and one approval flow traced through the code

Standard mode

The default preset

- C037 **Standard mode** — Default, the full tool set
- C265 **Tools** — Shell, files, web, skills, jobs
- C266 **Delegation** — Subagent, fork and workflow
- C267 **Planning and context** — Plans, goals, todos, compaction

PTC mode

Tools called from code

- C038 **PTC mode** — Tools called in batches from code
- C143 **Run code** — PTC mode, programs call the tools
- C268 **Standard tools** — Batched calls from generated code
- Off C269 **Workflow tool** — Switched off in this preset

Minimal mode

Only a terminal tool

- C039 **Minimal mode** — Only a terminal tool
- C144 **Persistent shell** — Minimal mode, one PTY terminal — bash, PowerShell
- C270 **Complete persona** — No runtime context added

Creator mode

Builds on DSH itself

- C040 **Creator mode** — Agent writes plugins for DSH
- C271 **Runtime inspection** — Read only view of the plugin API
- C272 **Plugin manager tool** — When a profile exists
- C273 **Creator skills** — Plugin and preset authoring help

The retry must be the exact same command, once, in the same turn, with a one sentence justification.

A child joins the parent's exact preset revision, then gets a delegation context and a tool filter.

Only the Lead creates or interrupts teammates.

A preset can be chosen only before the first turn of a session.

Subagent providers

One delegation seam, many providers

- C274 **Spawn** — A fresh child agent, in process
- C275 **Fork** — A child seeded from this history
- Off C165 **ACP subagent** — Drives an ACP agent as a child — ACP SDK 1.4.0
- Off C276 **Codex** — Official app server protocol — Codex 0.153.4
- Off C277 **Claude Code** — Official Agent SDK — Claude Agent SDK 0.3.263
- Off C278 **DSH SDK child** — Another dsh runtime over stdio

Continuation and limits

Children can keep going

- C279 **Active children** — Up to 8 at once
- C280 **Delegation depth** — One level by default
- C281 **Messages to a child** — Steer, wake or resume it
- C282 **Settle notice** — The parent hears when a child ends

Agent Teams

Experimental bundle

- C283 **Lead** — The root session leads the team
- C284 **Mailbox** — Durable team messages
- C285 **Task board** — Shared tasks with dependencies
- C286 **Teammates** — Up to 8 in the shipped bundle

Worked flow, a bash write that needs approval

Read left to right, top to bottom

1. C289 **Model calls bash** — The tool call is logged first
2. C290 **Pre execute passes** — No listener blocks the call
3. C291 **Confined run** — Sandbox in workspace write mode
4. C292 **Write denied** — Reported as a sandbox denial
5. C293 **Model asks for more** — Same command, a wider mode, a reason
6. C294 **Approval asked** — Logged, card shown to the user
7. C295 **Allowed once** — The decision is logged
8. C296 **Runs with wider access** — For this call only
9. C297 **Result logged** — The model continues the turn

What the flow writes

In the session log

- C104 **Tool calls and results** — Each call paired with its outcome
- C413 **Ask logged** — Before any channel answers
- C416 **Decision logged** — Paired with the ask by its id
- C298 **Denial in the result** — The model sees the sandbox message

Workflow engine

Detail on w3

- C024 **Workflow engine** — Scripts that run many subagents
- C287 **Script hooks** — Agent, parallel, pipeline, phase, log
- C288 **Runs in the code runtime** — Sandboxed Node process

Preset names are the ones the web UI shows. Off means switched off in the preset, or shipped but not composed into it.

Shipped — on by default

Off Shipped — off by default or opt in

Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

Lifecycle and operating model

How a running tree is composed and changed, and how the project is built and released

Composition order

Later layers win, row by row

1. **Bundle patches** C299 — In the profile bundle order
2. **Profile patch** C300 — Your layer for one profile
3. **Home patch** C301 — Applies to every profile
4. **Command line overlays** C302 — Patch files in argument order
5. **Telemetry switch** C303 — Applied last

Patch rules

- **Whole config replaced** C304 — A patch never merges into a row
- **Dump the tree** C305 — See composed rows before boot
- **Enable appends** C306 — A new bundle goes to the end

A patch replaces a row's whole config. It never merges into it.

Enabling a bundle appends it at the end of the list, which can change which layer wins.

Migrations never move, overwrite or delete a released session file.

Desktop clients only accept a greater version, so a feed cannot move an install backwards.

Plugin lifecycle

Install to unload, in order

1. **Install a bundle** C307 — pnpm add in the profile folder — pnpm 11.7.0
2. **Peer check** C420 — Rejects a plugin the runtime cannot satisfy
3. **Load into the tree** C421 — Loader imports and mounts it — Node 22.19 or 24
4. **Hot reload** C311 — Config changes apply without restart
5. **Unload undoes effects** C312 — Every registration is reversible

Install safeguards

What install guards against

- **Registry fallback** C308 — Tries the next registry
- **Restore on failure** C309 — Manifest and lock restored
- **Build script consent** C310 — Scripts wait for approval

Before a change merges

Repository practice

- **Pre push checks** C313 — Evidence matched to the change
- **Docs with every change** C314 — README and JSDoc updated together
- **Decision notes** C315 — Only for durable decisions
- **Bilingual docs** C316 — English and Chinese pairs checked

Release families

Four by workflow, desktop by an operator

- **npm packages** C317 — Tagged releases, manual publish
- **Vendored Cordis** C318 — Its own release family
- **Native addon** C319 — Its own release workflow
- **Python wheels** C320 — PyPI, and a GitLab registry build
- **Desktop installers** C321 — Packaged by an operator — electron updater 6.8.9

Desktop updates

Signed, one channel

- **Signed builds** C322 — Notarized macOS, signed Windows
- **Update feed** C323 — download.deepseek.com
- **Three targets** C324 — mac arm64, mac x64, win x64
- **Newer only** C325 — Clients only accept a greater version

Session format lifecycle

Released files are never rewritten

1. **Writer version** C326 — Bumped for structural changes
2. **Adjacent step** C327 — One package per version step
3. **Read opens** C328 — Migrate in memory only
4. **Write opens** C329 — Publish the next version beside it

Release blocks cite the workflow files and package scripts. Practices cite AGENTS.md or the gate script.

Shipped — on by default

Off Shipped — off by default or opt in

Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

Profile and bundle composition

What each shipped profile mounts, and what each web agent preset adds

	Web	Desktop	Headless	SDK	ACP	SDK minimal
C330 Agent loop	On	On	On	On	On	On
C331 DeepSeek adapter	On	On	On	On	On	On
C332 pi ai adapter	On	On	On	On	On	
C333 Session files	On	On	On	On	On	On
C334 Sandbox mode	Workspace write	Workspace write	Workspace write	Workspace write	Workspace write	Full access
C335 Approval	On	On	On	On	On	
C336 Shell tool	Per preset	Per preset	Platform gated	Platform gated	Platform gated	Platform gated
C337 File tools	Per preset	Per preset	On	On	On	
C338 Subagent tool	Per preset	Per preset	On	On	On	
C339 Compaction	Per preset	Per preset	On	On	On	
C340 Session telemetry	On	On	On	On	On	
C341 Session log in requests	On	On	On	On	On	On
C342 Runtime invariants						On
C343 Hot reload	With a profile	With a profile	Off	Off	Off	
C344 Surface	Web server	Web server	One shot runner	JSON RPC server	ACP server	JSON RPC server

	Standard	PTC	Minimal	Creator
C345 Shell tool	Platform gated	Platform gated	Platform gated	Platform gated
C346 File tools	On	On		On
C347 Web tools	On	On		On
C348 Subagent tool	On	On		On
C349 Workflow tool	On	Off		On
C350 Code mode		On		
C351 Plugin manager tool	Off	Off		With a profile

Empty cell means the profile does not mount that row. Detail on w6.

Every cell is read from the bundle patch files, composed in profile order.

In web and desktop, tools live in the agent presets, so those host rows read Per preset.

SDK minimal is one standalone bundle, and it pins full access.

Desktop reuses the web bundles under its own profile name.

The lower table shows the four web agent presets. Platform gated means bash or PowerShell by operating system.

The shell tool runs through bash sandbox, which tries bubblewrap first, then Landlock on Linux.

Shipped — on by default

Off Shipped — off by default or opt in

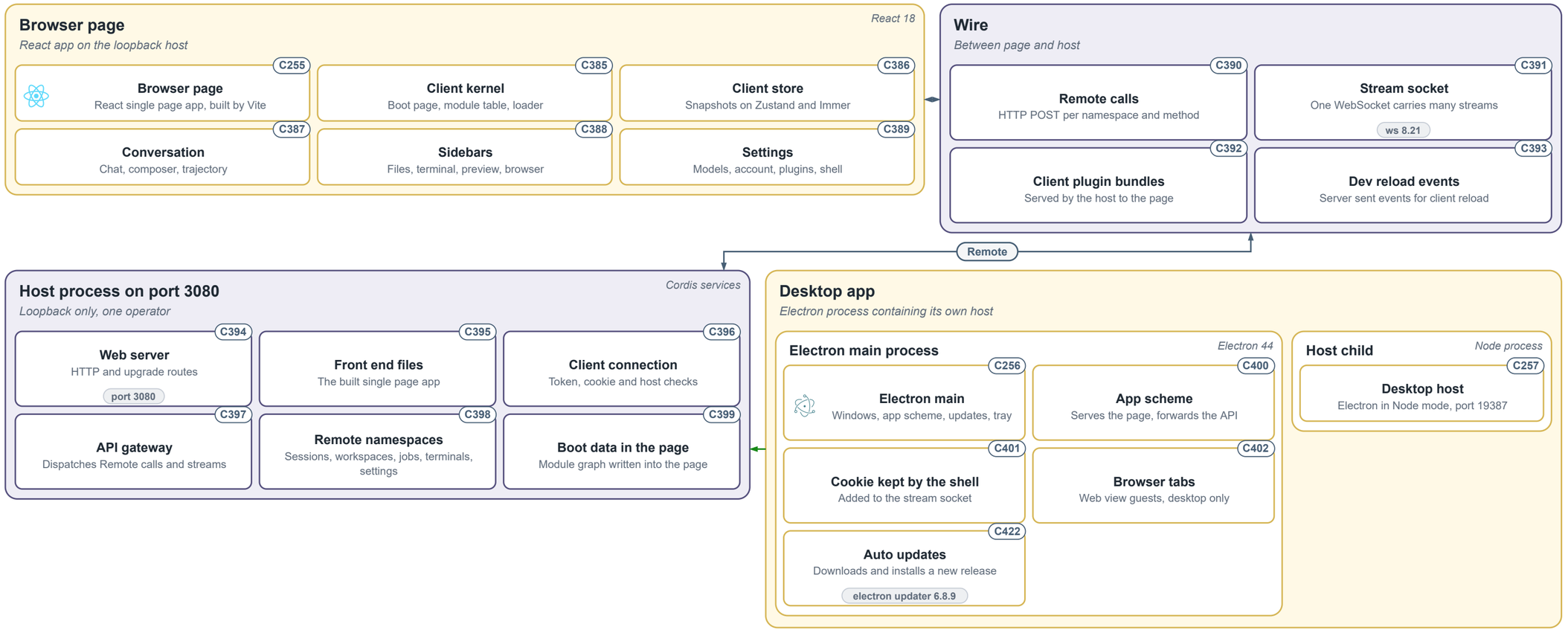
Experimental — experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP — protocols and trust links

Client and host topology

The browser page, the host process, the desktop shell and programs that drive dsh

Back to master | W10



Remote calls are HTTP POST requests and streams share one WebSocket. Server sent events serve only the development reload.

Desktop serves the page from its own scheme and forwards API calls to its host on port 19387.

The page authenticates with a signed HttpOnly cookie, issued against the launch token.

SDK and ACP clients spawn dsh and speak JSON RPC over its standard input and output.

The Electron main process checks for and installs updates itself, independent of the host child.

About sixty client packages ship. They are grouped here by what the user sees.

Shipped on by default

Off **Shipped** off by default or opt in

Experimental experimental package, opt in

MCP ACP JSON RPC Remote Trust OTLP protocols and trust links

Model layer

How a request reaches DeepSeek and back, and where the model is chosen

Model seam

Where every provider connects

- C423 **LLM seam** — One interface for every provider
- C424 **Credentials lookup** — Resolves the key before it is sent

Model request pipeline

One send, from build to log

1. C425 **Build request** — Turns become Messages fields
2. C426 **Resolve route** — Picks the provider and model
3. C427 **Send the request** — POST with the session id header
4. C428 **Stream the reply** — Decodes SSE with eventsource parser — eventsource parser 3.1
5. C429 **Retry on failure** — Empty response, rate limit, server, timeout, transport
6. C430 **Meter tokens** — Counts usage per session event
7. C431 **Append to session log** — Logged so the model call replays

Session history

DeepSeek endpoint

Reached over HTTPS

- C445 **DeepSeek API** — HTTPS, Anthropic Messages format

DeepSeek adapter

The default provider

- C432 **Official and account routes** — Same adapter, different credentials
- C433 **Anthropic Messages format** — DeepSeek speaks this wire format
- C434 **Wire extensions** — Plugins add fields, committed after send
- C435 **Default model catalog** — DeepSeek flash and V4 Pro — deepseek flash · deepseek v4 pro

pi ai adapter

OpenAI and Anthropic compatible

- C436 **pi ai library** — Provider catalog for LLM routes — pi ai 0.85.1
- C437 **OpenAI completions** — Chat completions route
- C438 **OpenAI responses** — Responses API route
- C439 **Anthropic messages** — Anthropic Messages route
- C440 **Self hosted endpoints** — OpenAI compatible, custom base URL

Where a model is chosen

Four independent picks

- C441 **Session default model** — Chosen when a session opens
- C442 **Subagent route default** — A child agent can pin its own
- C443 **Compaction model policy** — A policy can target another model
- C444 **Session title model** — A small model names the session

Every request carries the session id in a header and the turn history becomes its Messages fields.

Extension fields that plugins add are committed only after the response succeeds.

Retries cover rate limit, server, timeout and transport errors, not every failure.

The pi ai adapter is kept beside the DeepSeek adapter as a design verification twin.

A model key is resolved through credentials just before the request is sent.

Subagents may pin their own provider and model instead of inheriting the session default.

DeepSeek is the default provider. pi ai is kept beside it as a design verification twin.

Shipped — on by default · Off Shipped — off by default or opt in · Experimental — experimental package, opt in · MCP · ACP · JSON RPC · Remote · Trust · OTLP — protocols and trust links

Capability seam map

Each seam has a service definition, the providers that fill it, and the code that uses it

	Service definition	Mounted by a shipped profile	Other providers in the repo	Direct consumers
C352 **MCP resources** Serves MCP tool resources	mcp resources		mcp client	mcp resources
C353 **Browser use** Reserves the sole browser slot	browser use		browser use playwright mcp, browser use chrome devtools mcp, browser use stagehand native	browser use playwright mcp, browser use chrome devtools mcp, browser use stagehand native
C354 **Computer use** Reserves the sole computer slot	computer use		computer use cua driver mcp, computer use cua driver native	computer use cua driver mcp, computer use cua driver native
C355 **Attachments** Stores durable binary attachments	attachment	attachment local		api session controller, tool fs, llm pi ai and 1 more
C356 **Model service** Registry for streaming adapters	llm	llm deepseek, llm pi ai	llm replay	agent loop, compaction basic
C357 **DeepSeek request fields** Adds fields to API requests	deepseek llm api extensions	session log deepseek, plugin package inventory deepseek		llm deepseek
C358 **Session persistence** Persists session events as a log	session persistence	session persistence jsonl		agent loop, tool bash, hooks claude code and 4 more
C359 **Credentials** Resolves credential references	credentials	credentials local		api settings controller, llm deepseek, llm pi ai
C360 **DeepSeek account** Tracks account and tokens	deepseek account	deepseek account platform		api account controller, llm deepseek
C361 **Authorization** Runs flows to obtain credentials	authorization	authorization		llm pi ai, deepseek account platform
C362 **Session telemetry** Captures and redacts records	session telemetry	session telemetry otel		
C363 **Storage** Registers named storage backends	storage	storage json	storage sqlite	storage domain
C364 **Session query** Reads and searches history	session query	session query sqlite		session reference, tool session query
C365 **File references** Finds files for @ mentions	file reference	file reference local		api session controller
C366 **Session titles** Folds titles with a fallback	session title	session title first prompt llm	session title all prompts llm	
C367 **Questions to the user** Pauses tools for a human answer	user questions	user questions		tool ask user
C368 **Skills** Merges skill lists from providers	skill	skill filesystem, skill office	skill badge	tool skill

A seam is a swappable capability with three roles, a service definition, a provider and a consumer.

Rows come from the seam table in the repo docs. Mounted means a shipped profile or web preset loads it.

One provider swap changes the whole product, which is why these seams exist.

A logo on a row names its shipped default, for example the OpenTelemetry SDK or SQLite FTS5.

Package names are shown in plain words. Exact names are in the register.

Shipped on by default — **Shipped** (Off) off by default or opt in — **Experimental** experimental package, opt in — MCP, ACP, JSON RPC, Remote, Trust, OTLP protocols and trust links

Capability seam map, continued

Each seam has a service definition, the providers that fill it, and the code that uses it

	Service definition	Mounted by a shipped profile	Other providers in the repo	Direct consumers
C369 **Subprocess** Owns spawn and escalation	subprocess	subprocess local	subprocess ssh	bash local, bash sandbox, terminal bash and 4 more
C370 **Shell** Runs bash and PowerShell calls	shell	bash sandbox, pwsh sandbox	bash local, pwsh local	tool bash, tool pwsh, hooks claude code and 1 more
C371 **Terminals** Owns PTY session ids and cleanup	terminal	terminal bash		tool terminal
C372 **Sandbox** Confines argv via OS sandbox	sandbox	sandbox local	sandbox ssh	bash sandbox, terminal bash
C373 **Approval** Fails closed without an answerer	user approval	user approval		tools, tool bash, acp
C374 **Code runtime** Runs code on host bindings	ptc runtime	ptc runtime node	ptc runtime python	tools, workflow ptc
C375 **File system** Text IO plus atomic guarded edits	fs	fs sandbox	fs local, fs ssh	tool fs
C376 **Compaction** Summarizes history under pressure	compaction	compaction basic		compaction basic
C377 **Subagents** Registry of child transports	subagent	subagent spawn in process, subagent fork in process	subagent acp, subagent codex, subagent claude code and 1 more	tool subagent, tool subagent control, tool ralph
C378 **Speech to text** Routes audio to a recognizer	speech to text		speech to text sensevoice	api speech to text
C379 **Background jobs** Tracks background work by owner	jobs	jobs local		tool bash, tool pwsh, tool terminal and 3 more
C380 **Web search and fetch** Registers search and fetch	web	web search deepseek, web fetch http	web search exa, web search perplexity	tool web
C381 **Spill store** Saves oversized text as a locator	spill	spill local		spill policy
C382 **Directory picker** Opens a directory chooser	host directory picker	host directory picker native, host directory picker browse		api workspace controller
C383 **Workflow engine** Runs one engine per session	workflow	workflow ptc		tool workflow, tool ralph
C384 **Language servers** Normalizes four LSP query types	lsp		lsp stdio	tool lsp

Dashed cells list experimental providers only.

A capability seam is complete only with all three roles.

On Linux the sandbox tries bubblewrap first, then Landlock through a bundled native launcher.

pwsh sandbox also registers ctx.shell on Windows, a provider the repo docs table leaves out.

Package names are shown in plain words. Exact names are in the register.

Shipped on by default · Shipped (Off) off by default or opt in · Experimental experimental package, opt in · MCP ACP JSON RPC Remote Trust OTLP protocols and trust links

How to read these diagrams

One meaning per colour, one marker per maturity state

Maturity markers

Shipped	On by default in that shipped profile
Off (Off)	Shipped, but off by default or opt in
Experimental	Package under the experimental group

Layer colours

The same meaning on every page

- **A** Execution world
- **B** External services
- **C** Protocols and API
- **D** Session data
- **E** Model layer
- **F** Harness core
- **G** Security and observability
- **H** People and surfaces

Links and protocols

MCP	An agent calling a tool on an MCP server	ACP	An editor or IDE driving the harness as an agent
JSON RPC	The protocol the Python and TypeScript SDKs use	Remote	A generated Cordis Remote call to the host
Trust	An authentication or credential check link	OTLP	Session telemetry exported to a collector

Line styles

Only the kinds this set actually draws

- A plain call or a data hand off
- A live tool call over MCP
- A two way ACP client session
- An SDK request and its reply
- A two way Cordis Remote call
- An authentication or credential check

Chips and badges

C001	Off	SQLite	*Cordis plugins*	1
Points at a row in the citation register	Rides on a shipped block that is off by default	Names the exact system that block uses	Names a zone main technology, top right	Marks one step in an ordered sequence

Citation register

444 citations in the set

Every citation id in this set, in id order

Id	Item	Pages	Source
C001	Web app	e0,w0	packages/bundle/web-app
C002	Desktop app	e0,w0	apps/desktop
C003	One shot command line	e0,w0	packages/bundle/headless
C004	SDK programs	e0,w0	packages/sdk/client
C005	Agent loop	e0,w0	packages/core/agent-loop
C006	Tool registry	e0,w0	packages/core/tools
C007	Subagents	e0,w0	packages/subagent/subagent
C008	Session log	e0,w0	packages/core/session
C009	Remote over SSH	e0,w0	packages/ssh/ssh
C010	MCP client	e0,w0	packages/mcp/mcp-client
C011	DeepSeek API	e0,w0	packages/llm/llm-deepseek-api-key
C012	Compatible endpoints	e0,w0	packages/llm/llm-pi-ai
C013	Web search providers	e0,w0	packages/web/web-search-deepseek
C014	Sandbox	e0,w0	packages/sandbox/sandbox-local
C015	Approval	e0,w0	packages/interaction/user-approval
C016	Every step in the session log	e0	packages/core/session
C017	ACP clients	w0	packages/acp/acp
C018	Plugin tree	w0	packages/boot/app-boot
C019	Web server and API gateway	w0	packages/api/gateway
C020	Agent registry	w0	packages/core/agent
C021	System prompt	w0	packages/core/system-prompt
C022	Model selection	w0	packages/core/agent-default-model
C023	Context compaction	w0	packages/compaction/compaction-basic
C024	Workflow engine	w0	packages/workflow/workflow-ptc
C025	Plan, goals and todos	w0	packages/plan/plan-mode
C026	Session search	w0	packages/session-query/session-query-sqlite
C027	Skills and instructions	w0	packages/skill/skill-filesystem
C028	Shell	w0	packages/shell/bash-sandbox
C029	Subprocesses	w0	packages/subprocess/subprocess-local
C030	File system	w0	packages/fs/fs-sandbox
C031	Code runtime	w0	packages/ptc-runtime/ptc-runtime-node

413 more citations are listed in citation_register.csv

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